

Academic Organization Ontology

Student: Syeda Khadija Batool Gardezi
Università degli studi di Verona

1. Introduction

For this project, I created a small academic organization ontology using Protégé and OWL. The main idea was to represent some common parts of a university, such as students, professors, courses, departments, and research projects, and show how they are connected to each other.

2. Class Hierarchy

I started by organizing the main concepts into a class hierarchy. I used three main classes: **Person**, **AcademicUnit**, and **AcademicActivity**.

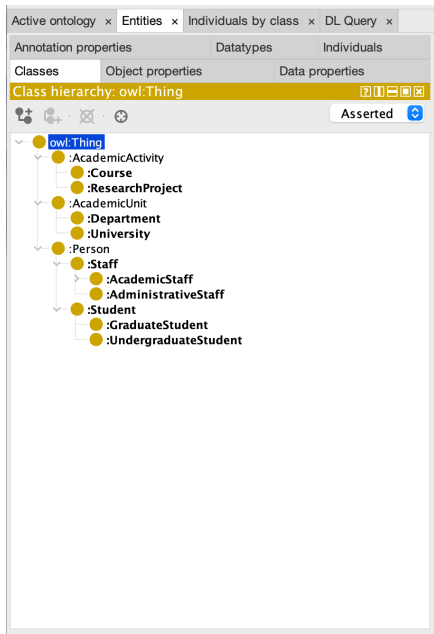
The **Person** class is divided into students and staff. Students can be graduate or undergraduate students, while staff includes academic and administrative staff. Under academic staff, I added professors and lecturers.

The ontology is organized into three main categories under **owl:Thing**:

owl:Thing

- AcademicActivity
 - Course
 - ResearchProject
- AcademicUnit
 - Department
 - University
- Person
 - Staff
 - AcademicStaff
 - Lecturer
 - Professor
 - AdministrativeStaff
 - Student
 - GraduateStudent
 - UndergraduateStudent

This hierarchy represents relationships between academic concepts. For example, a GraduateStudent is a type of Student, and a Student is a type of Person. Similarly, a Professor is a type of AcademicStaff.



3. Object Properties

Object properties are used to define relationships between individuals in the ontology.

The ontology contains the following object properties:

- enrolledIn
- hasDepartment
- participatesIn
- studiesIn
- supervises
- teaches
- worksIn

These properties allow relationships between different entities to be represented.

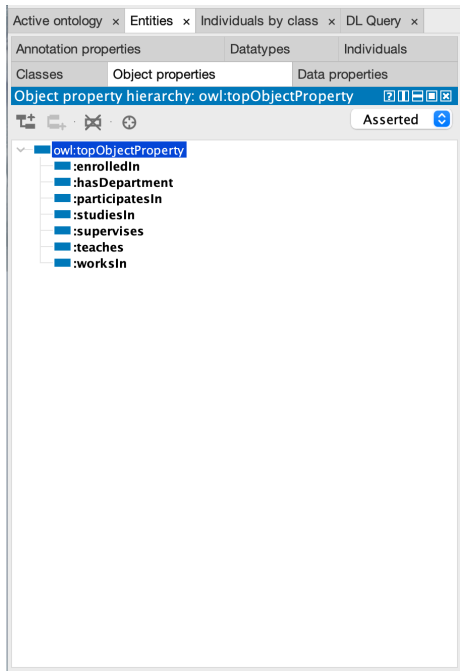
Examples include:

KhadijaGardezi → **enrolledIn** → **KnowledgeRepresentation**

KhadijaGardezi → **studiesIn** → **ComputerScienceDepartment**

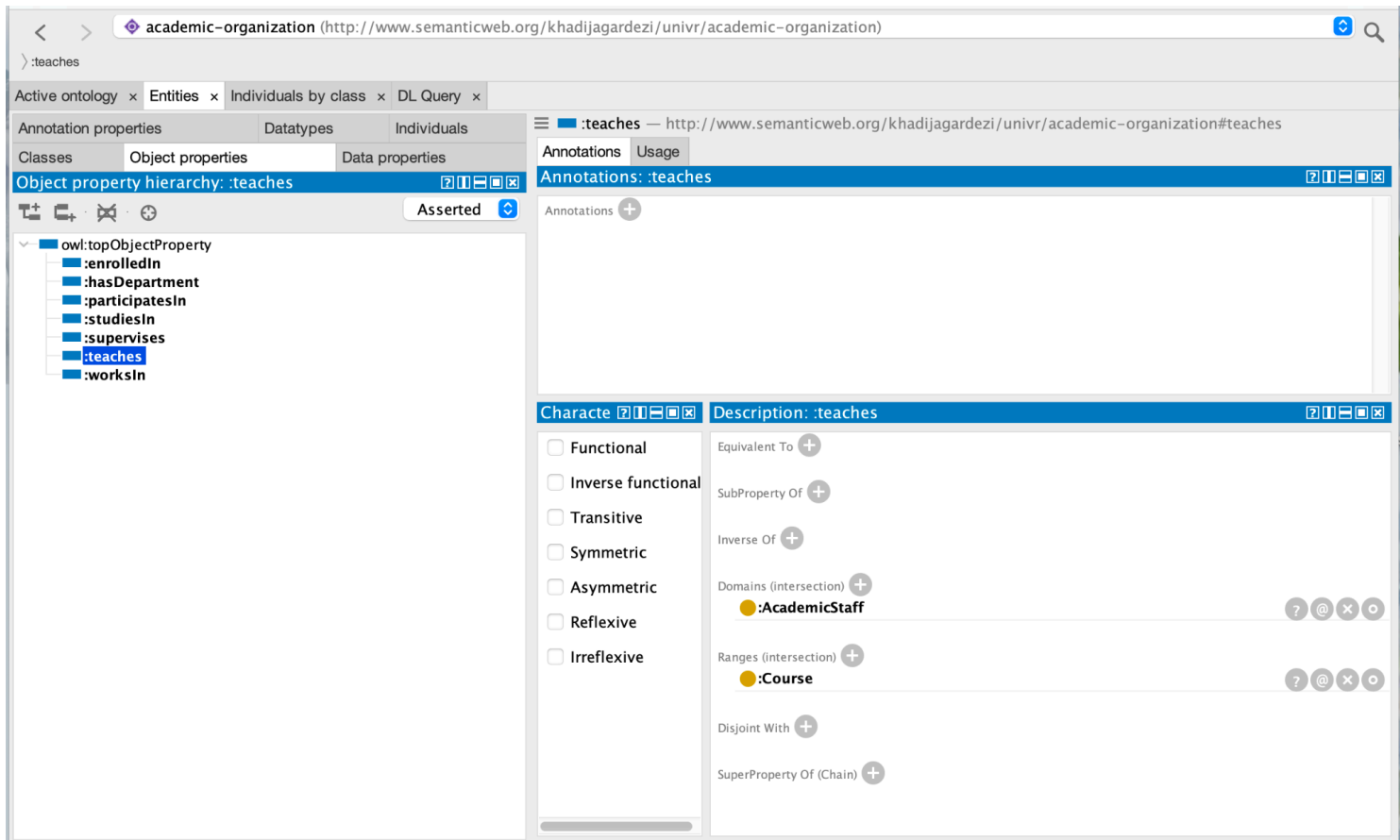
MatteoCristani → **teaches** → **KnowledgeRepresentation**

MatteoCristani → **worksIn** → **ComputerScienceDepartment**



4. Domain and Range of the teaches Property

The **teaches** object property represents the relationship between academic staff members and courses.



5. Individuals

This represents concrete entities in the academic environment.

The main individuals created for this model are:

KhadijaGardezi
MatteoCristani
KnowledgeRepresentation
ComputerScienceDepartment
UniversityOfVerona

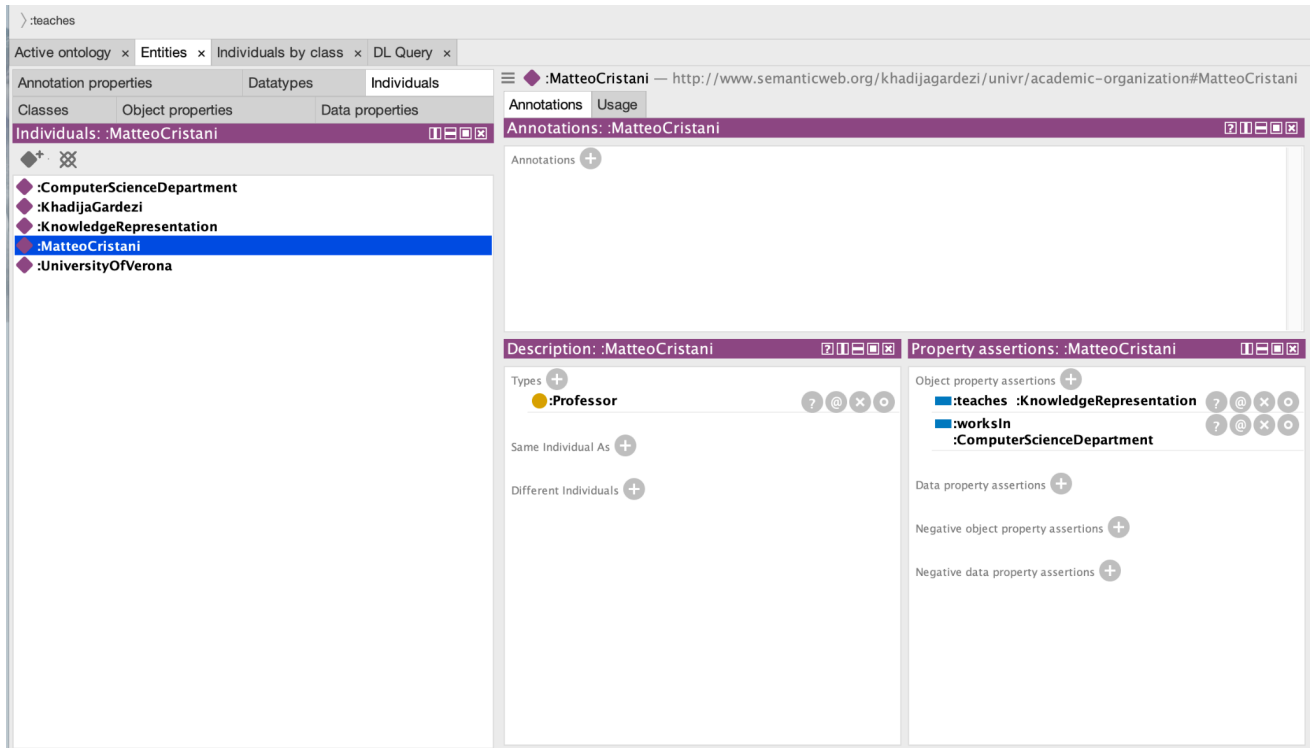
These individuals provide a small example of how students, professors, courses, departments, and universities can be represented and connected within the ontology.

6. Professor Individual

MatteoCristani is defined as an instance of the **Professor** class.

The ontology contains the following relationships for this individual:

MatteoCristani → teaches → KnowledgeRepresentation | MatteoCristani → worksIn → ComputerScienceDepartment

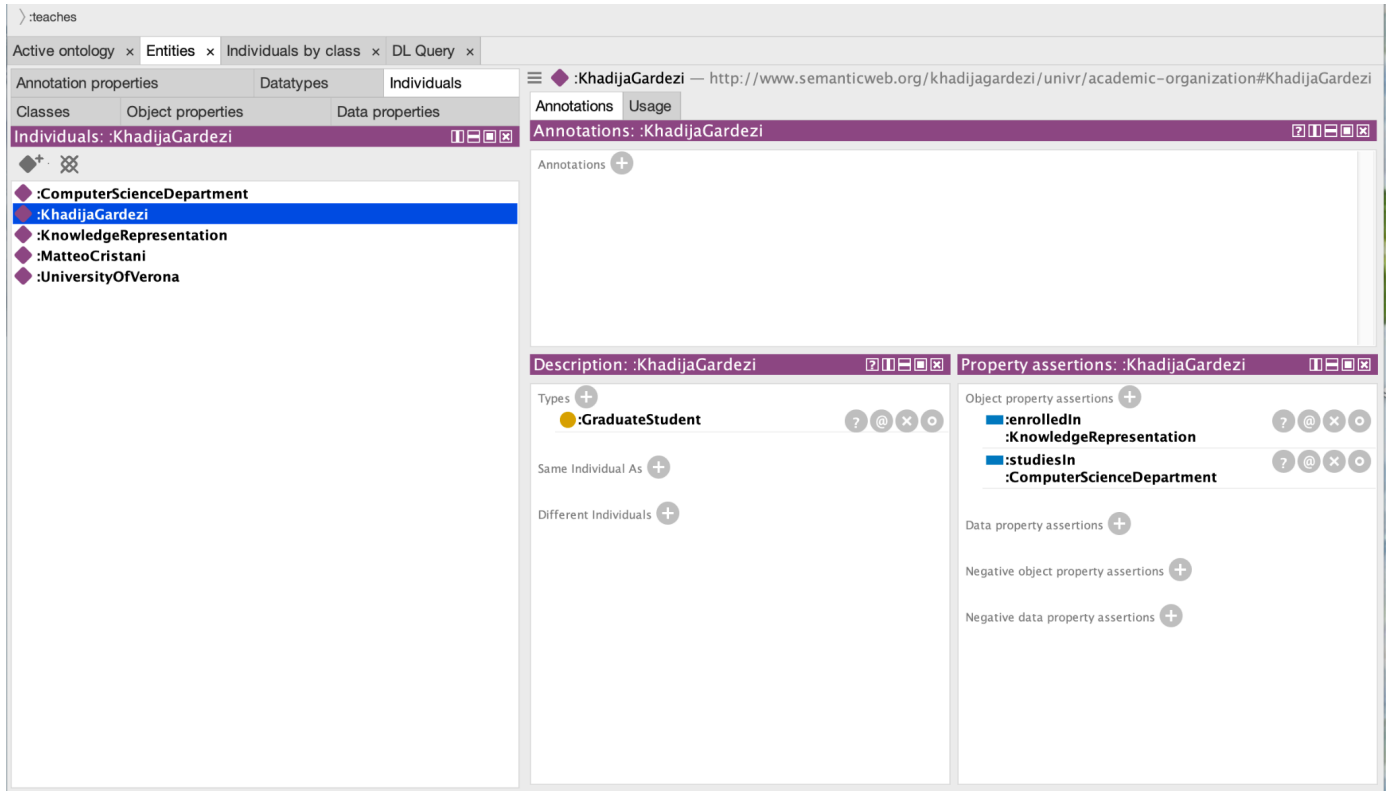


7. Graduate Student Individual

KhadijaGardezi is defined as an instance of the GraduateStudent class.

The following relationships are associated with this individual:

KhadijaGardezi → enrolledIn → KnowledgeRepresentation | KhadijaGardezi → studiesIn → ComputerScienceDepartment



8. OWL Restrictions

For the GraduateStudent class, the ontology contains: GraduateStudent → SubClassOf → Student

And GraduateStudent → SubClassOf → enrolledIn some Course

The expression enrolledIn some Course is an existential restriction.

For the Professor class, the ontology contains: Professor → SubClassOf → AcademicStaff

And Professor → SubClassOf → teaches some Course

The expression teaches some Course states that a Professor must have at least one teaches relationship with a Course.

9. Disjoint Classes

The ontology defines Student and Staff as disjoint classes:

Student → DisjointWith → Staff

This means that the two classes are mutually exclusive.

10. Reasoning with Hermit

The Hermit 1.4.3.456 reasoner was used to perform reasoning over the ontology.

In this project, reasoning considers information such as:

- Class and subclass relationships
- Object property domains and ranges
- OWL restrictions
- Disjoint classes
- Individual types
- Object property assertions

GraduateStudent is a subclass of **Student**, an individual defined as a GraduateStudent can also be recognized as a Student.

Similarly, a **Professor** is a subclass of **AcademicStaff**, a Professor is also recognized as AcademicStaff.

11. DL Query

This functionality was used to test the ontology.

The following query was executed:

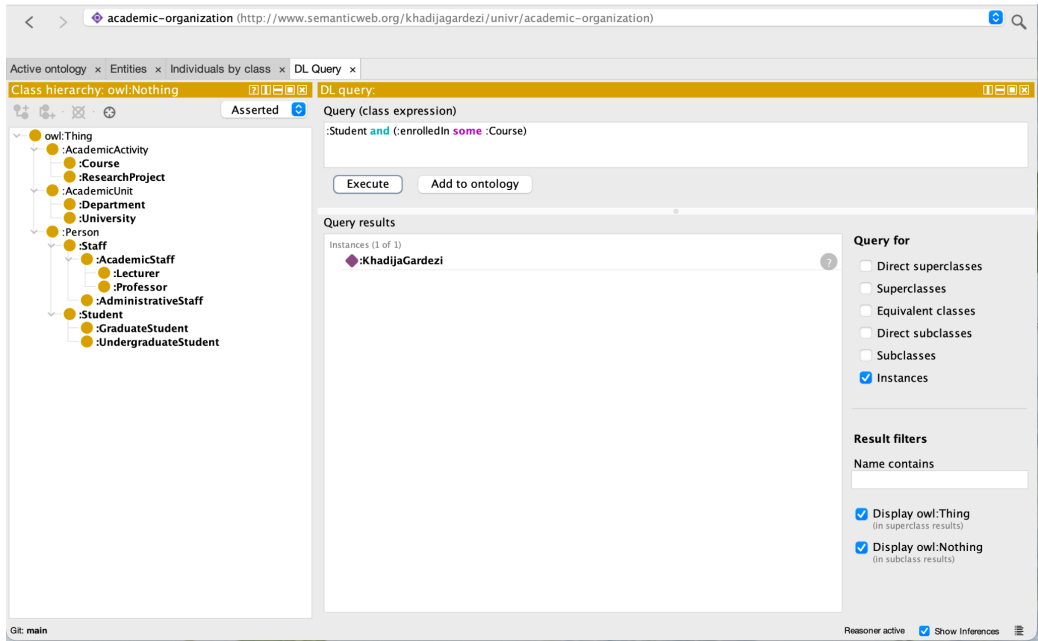
:Student and (:enrolledIn some :Course)

The query asks:

Which individuals are Students and are enrolled in at least one Course?

With **Instances** selected in the DL Query interface, the result was:

KhadijaGardezi



12. Conclusion

The Academic Organization Ontology demonstrates the basic principles of ontology engineering using OWL and Protégé.

The HermiT reasoner was used to reason over the ontology and evaluate its logical structure. In particular, the query for students enrolled in a course successfully returned.